

PDVA 502 CPG

NOT
GNF

$$\begin{array}{l} \Rightarrow asb \\ \Rightarrow ob \end{array}$$

i.e., $W = \phi^N B$.
first find the

story to this L or Grouse
aabb

Rules

- ① $\delta(q, \epsilon, \epsilon) \rightarrow (q, a, b)$
- ② $\delta(q, \epsilon, \epsilon) \rightarrow (q, a, b)$
- ③ $\delta(q, a, a) \rightarrow (q, \epsilon)$
- ④ $\delta(q, b, b) \rightarrow (q, \epsilon)$

010000	S
010000	0BB
10000	BB
10000	1SB
0000	SB
0000	0BBB
0000	BBB
000	0BB
00	BB
00	0B
0	B
0	0
When in	GNF
ϵ	ϵ

[illegible]

When in GNF

~~string can be 000, 00010,~~

$$\begin{aligned} S &\rightarrow OBB \\ B &\rightarrow OS|IS|O \end{aligned}$$

Rules

- ① $\delta(q, 0, 5) \rightarrow (q, BB)$
- ② $\delta(q, 0, B) \rightarrow (q, 5)$
- ③ $\delta(q, 1, B) \rightarrow (q, 5)$
- ④ $\delta(q, 0, B) \rightarrow (q, \epsilon)$

$q_1, \epsilon, S) \rightarrow (q, 0BB)$
 $\delta(q, \epsilon, B) \rightarrow (q, 0S)$
 $\delta(q, \epsilon, B) \rightarrow (q, 1S)$
 $\delta(q, \epsilon, B) \rightarrow (q, 0)$
 $(q, 1, 1) \rightarrow (q, \epsilon)$
 $(q, 0, 0) \rightarrow (q, \epsilon)$

Take string



5000000 S
 10000 BB
 0000 SA
 000 BBB
 00 BBB
 0 BBB
 0 BBB

0	0000	3
0	0000	0BB
1	0000	B B
1	0000	15 B
0	0000	S B
0	0000	09BB
0	00	8BB
0	00	0BB
0	00	0BB
0	00	0B
0	0	0B

Construction of PDA equivalence of CFG

Rules

- CFG $\Rightarrow$ PDA
- ① $S \rightarrow \epsilon$ $\delta(P, \epsilon, \epsilon) \rightarrow (Q, S)$ Initial state
 $\delta(P, \epsilon, S) \rightarrow (Q, \epsilon)$
 - ② $A \rightarrow x$ $\delta(Q, \epsilon, A) \rightarrow (Q, x)$
 - ③ $a \in V_t$ $\delta(Q, a, a) \rightarrow (Q, \epsilon)$
 if input is a & top of stack is a then pop
 $Q \in V_t$
- Design PDA for CFG

- ① $S \rightarrow \epsilon$
- ② $S \rightarrow SS$
- ③ $S \rightarrow (S)$

first find the string belong to this CFG
 $() () () ((()))$

$$P = (Q, \Sigma, \Gamma, \delta, S, F)$$

$$Q \rightarrow (P, Q) (P, Q)$$

$$\Sigma \rightarrow \{ (,) \}$$

$$S \rightarrow P \quad \text{PDA} (Q, \Sigma, \delta, q_0, F)$$

$$F \ni Q \quad \text{PDA} (Q, \Sigma, \delta, q_0, F, \Gamma, z_0)$$

$$\textcircled{1} \delta(P, \epsilon, \epsilon) \rightarrow (Q, S) \quad \text{— initial state}$$

$$\textcircled{2} \delta(Q, \epsilon, S) \rightarrow (Q, \epsilon) \quad \text{— ①}$$

$$\textcircled{3} \delta(Q, \epsilon, S) \rightarrow (Q, SS) \quad \text{— ②}$$

$$\textcircled{4} \delta(Q, \epsilon, S) \rightarrow (Q, (S)) \quad \text{— ③}$$

$$\textcircled{5} \delta(Q, (, (S)) \rightarrow (Q, \epsilon) \quad \text{— terminal}$$

$$\textcircled{6} \delta(Q,), (S)) \rightarrow (Q, \epsilon) \quad \text{— terminal}$$

Design PDA for CFG

$S \rightarrow aSa$
 $S \rightarrow bSb$
 $S \rightarrow c$

Soln

PDA

$P = (Q, \Sigma, T, \delta, S, F)$
 $Q = \{P, Q\}$

$\Sigma = \{a, b, c\}$
 $T = \{S, a, b, c\}$
 $S = P$
 $F = \{Q\}$

Rules

- ① $\delta(P, \epsilon, \epsilon) \rightarrow (P, \epsilon)(Q, S)$
- ② $\delta(Q, \epsilon, S) \rightarrow (Q, aSa)$
- ③ $\delta(Q, \epsilon, S) \rightarrow (Q, bSb)$
- ④ $\delta(Q, \epsilon, S) \rightarrow (Q, c)$
- ⑤ $\delta(Q, \epsilon, S) \rightarrow (Q, \epsilon)$
- ⑥ $\delta(Q, a, a) \rightarrow (Q, \epsilon)$
- ⑦ $\delta(Q, b, b) \rightarrow (Q, \epsilon)$
- ⑧ $\delta(Q, c, c) \rightarrow (Q, \epsilon)$

Let's take $w = abbcbbba$

S.No.	State	Input	Stack	Rule
		abbcbbba	ϵ	
		a b b c b b a	S	
		a b b c b b a	a S a	
		a b b c b b a	S a	
		b b c b b a	b S b a	
		b b c b b a	S b a	
		b b c b b a	b S b b a	
		b c b b a	S b b a	
		b c b b a	c b b a	
		c b b a	b b a	
		c b b a	b a	
		c b a	a	
		c b a		

follows

$$\begin{aligned} S &\rightarrow 0B \mid 1A \\ A &\rightarrow 0S \mid 1AA \mid 0 \\ B &\rightarrow 1S \mid 0BB \mid 1 \end{aligned}$$

Design PDA M corresponding to given CFG.

Soln

We define PDA

$$M = \{Q, \Sigma, \Gamma, \delta, q_0, z_0, F\}$$

where $Q = \{q_0, q_1, q_f\}$

$$\Sigma = \{0, 1\}$$

$$\Gamma = \{0, 1, S, A, B, z_0\}$$

$q_0 =$ start state.

$z_0 =$ start stack symbol.

$F \rightarrow$ final state q_f .

δ can be as follows.

~~$$\delta(q_0, \epsilon, S) \rightarrow (q_1, 0B)$$~~

① $\delta(q_0, \epsilon, z_0) \rightarrow (q_1, Sz_0)$

i.e. initial stack symbol.

② $\delta(q_1, \epsilon, S) \rightarrow (q_1, 0B)$

i.e. Replace Variable using Production.

$$\delta(q_1, \epsilon, S) \rightarrow (q_1, 1A)$$

$$\delta(q_1, \epsilon, A) \rightarrow (q_1, 0S)$$

$$\delta(q_1, \epsilon, A) \rightarrow (q_1, 1AA)$$

$$\delta(q_1, \epsilon, A) \rightarrow (q_1, 0)$$

$$\delta(q_1, \epsilon, B) \rightarrow (q_1, 1S)$$

$$\delta(q_1, \epsilon, B) \rightarrow (q_1, 0BB)$$

$$\delta(q_1, \epsilon, B) \rightarrow (q_1, 1)$$

③ match terminator

$$\delta(q_1, 0, 0) \rightarrow (q_f, \epsilon)$$

$$\delta(q_1, 1, 1) \rightarrow (q_f, \epsilon)$$

④ accept condition.

$$\delta(q_1, \epsilon, z_0) \rightarrow (q_f, z_0)$$

Consider string 1010

1010 1A.

010 A.

000 0S

10 S.

10 1A.

0 A.

00

$\epsilon \epsilon$

find a PDA equivalent to the grammar
 $S \rightarrow aSbb|a$

Soln Simulate the designed PDA for the string 'aabb'
 we define PDA

$M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, f)$
 where $Q = \{q_0, q_1, q_f\}$ $\Gamma = \{a, b, S, z_0\}$
 $\Sigma = \{a, b\}$ $q_0 = \text{initial state}$ $q_f \rightarrow \text{final state}$

$z_0 = \text{initial symbol}$
 $q_f \rightarrow \text{final state}$

- (1) Initial stack $\delta(q_0, \epsilon, z_0) \rightarrow (q_1, Sz_0)$
- (2) Replacement Variable
 $\delta(q_1, \epsilon, S) \rightarrow (q_1, aSbb)$
 $\delta(q_1, \epsilon, S) \rightarrow (q_1, a)$
- (3) match remaining $\delta(q_1, a, a) \rightarrow (q_1, \epsilon)$
 $\delta(q_1, b, b) \rightarrow (q_1, \epsilon)$
- (4) Accept Condition $\delta(q_1, \epsilon, z_0) \rightarrow (q_f, z_0)$

Now consider string "aabb"

stack	input	stack	
q_0	aabb	z_0	
q_1	aabb	Sz_0	push S
q_1	aabb	$aSbbz_0$	match a.
q_1	abb	$sbbz_0$	$s \rightarrow a$
q_1	<u>a</u> bb	<u>a</u> bbz ₀	match a.
q_1	<u>b</u> b	<u>b</u> bbz ₀	match b.
q_1	<u>b</u>	<u>b</u> z ₀	match b.
q_1	<u>ε</u>	<u>z</u> ₀	Accept.
q_f	<u>ε</u>	<u>z</u> ₀	Accept.

Find a PDA equivalent to the given grammar.

$S \rightarrow XY$ i.e. generate

$X \rightarrow aX/\epsilon$ i.e. generate a^n

$Y \rightarrow bYc/\epsilon$ i.e. generate $b^m c^m$.

Soln.

So our language will be -

$$L = \{a^n b^m c^m \mid n \geq 0, m \geq 0\}.$$

PDA

$$M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$$

where

$$Q = \{q_0, q_1, q_f\}$$

$q_0 \rightarrow$ start state

$$\Sigma = \{a, b, c\}$$

$Z_0 \rightarrow$ initial symbol on stack

$$\Gamma = \{a, b, c, S, X, Y, Z_0\} \quad q_f \rightarrow \text{final state}$$

δ proc

① initial stack $\delta(q_0, \epsilon, Z_0) \rightarrow (q_1, SZ_0)$

② replacement

$$\begin{aligned} \delta(q_1, a, S) &\rightarrow (q_1, X, Y) \\ \delta(q_1, a, X) &\rightarrow (q_1, aX) \\ \delta(q_1, \epsilon, X) &\rightarrow (q_1, \epsilon) \\ \delta(q_1, \epsilon, Y) &\rightarrow (q_1, bYc) \\ \delta(q_1, \epsilon, Y) &\rightarrow (q_1, \epsilon) \end{aligned}$$

③ match terminated

$$\begin{aligned} \delta(q_1, a, a) &\rightarrow (q_1, \epsilon) \\ \delta(q_1, b, b) &\rightarrow (q_1, \epsilon) \\ \delta(q_1, c, c) &\rightarrow (q_1, \epsilon) \end{aligned}$$

④ Accept condition $\delta(q_1, \epsilon, Z_0) \rightarrow (q_f, Z_0)$

Consider string aa bbbccc.

State	input	stack
q_0	aa bbbccc	Z_0
q_1	aa bbbccc	SZ_0
q_1	a bbbccc	XYZ_0
q_1	a bbbccc	$aXYZ_0$
q_1	a bbbccc	$aXYZ_0$
q_1	a bbbccc	$aXYZ_0$
q_1	a bbbccc	$aXYZ_0$
q_1	bbbccc	$aXYZ_0$

bbbccc	YZ_0
bbbccc	$bYcZ_0$
bbccc	YcZ_0
bbccc	$bYccZ_0$
bccc	$YccZ_0$
bccc	$bYcccZ_0$
ccc	$YcccZ_0$
ccc	$cccZ_0$
cc	ccZ_0

$\epsilon \epsilon Z_0$
 ϵZ_0
 accepted

Convert PDA to CFG

$M = \{P, Q\}, \{0, 1\}, \{x, z\} \delta, q, z$
 $Q \subseteq$ transition initial final
 δ transition symbol on stack

- ① $\delta(q, 1, z) \rightarrow (q, xz)$ Push
- ② $\delta(q, 1, x) \rightarrow (q, xx)$ Push
- ③ $\delta(q, \epsilon, x) \rightarrow (q, \epsilon)$ Pop
- ④ $\delta(q, 0, x) \rightarrow (p, x)$ NOP
- ⑤ $\delta(p, 1, x) \rightarrow (p, \epsilon)$ Pop
- ⑥ $\delta(p, 0, z) \rightarrow (q, z)$ NOP

qzq	$qzzs$
qzp	qzp
pzz	pzz
pzp	pzp

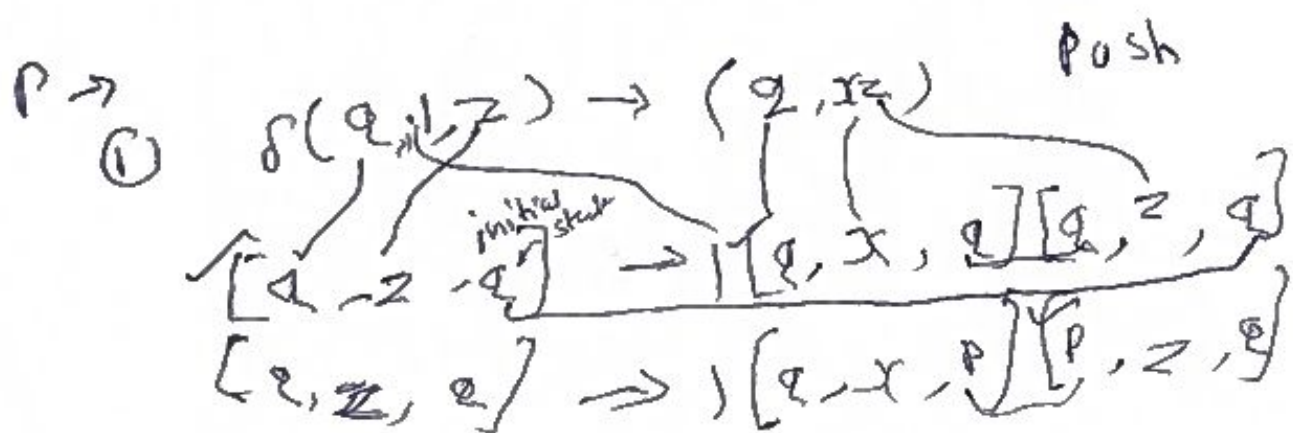
$T = \{0, 1\}$

$S \rightarrow (q, z, q)$
 $S \rightarrow (q, z, p)$

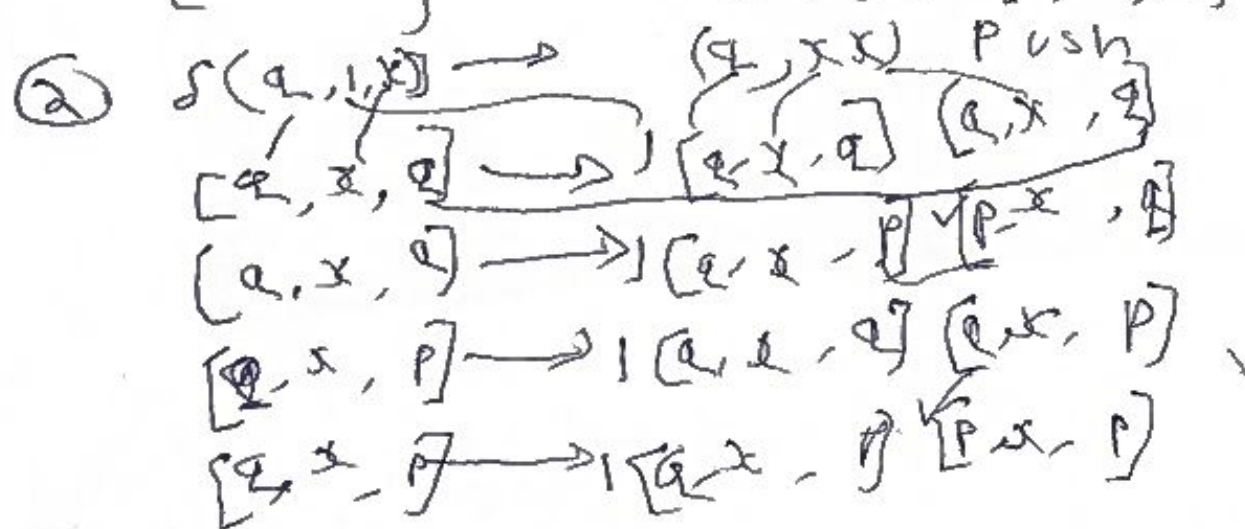
initial state initial symbol

$z = 4$

4-row 2-column - at R.T.S.



$[q, z, p] \rightarrow [q, x, q] [q, z, p]$
 $[q, z, p] \rightarrow [q, x, p] [p, z, p]$



$$\delta(q, \varepsilon, x) \rightarrow (q, \varepsilon) \text{ POP}$$

$$[q, x, q] \rightarrow \varepsilon$$

No operation

$$\begin{aligned} \textcircled{4} \quad \delta(q, 0, x) &\rightarrow (p, x) \\ [q, x, q] &\rightarrow 0[p, x, q] \\ [q, x, p] &\rightarrow 0[p, x, p] \end{aligned}$$

$q^1 = 2$ Row
1 - Column

$$\begin{aligned} \textcircled{5} \quad \delta(p, 1, x) &\rightarrow (p, \varepsilon) \text{ POP} \\ [p, x, p] &\rightarrow 1 \end{aligned}$$

$q^0 = 1$ Row
1 - column of P.H.S

$$\begin{aligned} \textcircled{6} \quad \delta(p, 0, z) &\rightarrow (q, z) \text{ No operation PDA} \\ [p, z, q] &\rightarrow 0[q, z, q] \\ [p, z, p] &\rightarrow 0[q, z, p] \end{aligned}$$

Production

Replace each Production by Same Symbol.

$$\begin{aligned} [q, z, q] &\rightarrow A & [q, x, q] &\rightarrow E \\ [q, z, p] &\rightarrow B & [q, x, p] &\rightarrow F \\ [p, z, q] &\rightarrow C & [p, x, q] &\rightarrow G \\ [p, z, p] &\rightarrow D & [p, x, p] &\rightarrow H \end{aligned}$$

$$S \rightarrow A$$

$$S \rightarrow B$$

$$A \rightarrow 1EA$$

$$A \rightarrow 1FC$$

$$B \rightarrow 1EB$$

$$B \rightarrow 1FD$$

$$E \rightarrow 1EE \quad E \rightarrow \varepsilon$$

$$E \rightarrow 1FG \quad E \rightarrow 0G$$

$$F \rightarrow 1EF \quad F \rightarrow 0H$$

$$F \rightarrow 1FH \quad H \rightarrow 1$$

$$C \rightarrow 0A$$

$$b \rightarrow 0b$$

$$M = (\underbrace{\{q_0, q_1\}}_Q, \underbrace{\{a, b\}}_\Sigma, \underbrace{\{a, z_0\}}_\Gamma, \delta, \underbrace{q_0}_{\text{initial state}}, \underbrace{z_0}_{\text{initial symbol on stack}}, \underbrace{\emptyset}_F)$$

Final state

$$\textcircled{1} \delta(q_0, a, z_0) = (\cancel{q_0, a, z_0}) (q_0, az_0)$$

$$\textcircled{2} \delta(q_0, a, a) = (\cancel{q_0, a, a}) (q_0, aa)$$

$$\textcircled{3} \delta(q_0, b, a) \rightarrow (q_1, a)$$

~~$$\textcircled{4} \delta(q_1, b, a) \rightarrow (q_1, \epsilon)$$~~

~~$$\textcircled{4} \delta(q_1, b, a) \rightarrow (q_1, a)$$~~

~~$$\textcircled{5} \delta(q_1, a, a) \rightarrow (q_1, \epsilon)$$~~

$$\textcircled{5} \delta(q_1, \epsilon, z_0) \rightarrow (q_1, \epsilon)$$

$$\textcircled{6} \delta(q_1, \epsilon, z_0) \rightarrow (q_1, \epsilon)$$

$$Sol^n \rightarrow G = (V, T, P, S)$$

$$V = (S, [q_0, a, z_0], [q_0, a, a], [q_1, az_0], [q_1, a, a], [a_0, z_0, a_0], [q_0, z_0, a], [q_1, z_0, a], [q_1, z_0, \epsilon])$$

$$T \rightarrow \{a, b\}$$

$$S \rightarrow [q_0, z_0, z_0]$$

$$S \rightarrow [a_0, z_0, a]$$

initial state

$$\delta(q_0, 0, z_0) \rightarrow (q_0, a_0, z_0)$$

$$[q_0, z_0, q_0] \rightarrow a[q_0, a, q_0] [q_0, z_0, q_0]$$

$$[q_0, z_0, q_0] \rightarrow a[q_0, a, q_1] [q_0, z_0, q_0]$$

$$[q_0, z_0, q_1] \rightarrow a[q_0, a, q_0] [q_0, z_0, q_1]$$

$$[q_0, z_0, q_1] \rightarrow a[q_0, a, q_1] [q_0, z_0, q_1]$$

$$(2) \delta(q_0, a, a) \rightarrow (q_0, 0, \underline{a})$$

$$[q_0, a, q_0] \rightarrow a[q_0, a, q_0] [q_0, a, q_0]$$

$$[q_0, a, q_0] \rightarrow a[q_0, a, q_1] [q_0, a, q_0]$$

$$[q_0, a, q_1] \rightarrow a[q_0, a, q_0] [q_0, a, q_1]$$

$$[q_0, a, q_1] \rightarrow a[q_0, a, q_1] [q_0, a, q_1]$$

$$(3) \delta(q_0, b, 0) \rightarrow (q_1, 0)$$

$$[q_0, b, q_0] \rightarrow b[q_1, 0, q_0]$$

$$[q_1, 0, q_0] \rightarrow b[q_1, 0, q_0]$$

$$(4) \delta(q_1, b, 0) \rightarrow (q_1, a)$$

$$[q_1, a, q_0] \rightarrow b[q_1, a, q_0]$$

$$[q_1, a, q_0] \rightarrow b[q_1, a, q_0]$$

$$(5) \delta(q_1, a, a) \rightarrow (q_1, \varepsilon)$$

$$[q_1, 0, q_0] \rightarrow \varepsilon$$

(6)

$$\delta(q_1, \varepsilon, z_0) \rightarrow (q_1, \varepsilon)$$

$$(q_1, z_0, q_1) \rightarrow \varepsilon$$

6.0m

$Q_0, 20, 20 \rightarrow 2$

$Q \rightarrow 0.62$

$Q_0, 20, 20 \rightarrow 2$

$Q \rightarrow 0.62$

$Q \rightarrow 0.62$

$Q_0, 0, 20 \rightarrow 0$

$Q \rightarrow 0.62$

$Q_0, 20, 20 \rightarrow 0$

$Q \rightarrow 0.62$

$Q_0, 0, 20 \rightarrow 0$

$Q \rightarrow 0.62$

$Q_0, 20, 20 \rightarrow 0$

$Q \rightarrow 0.62$

$Q \rightarrow 0.62$

$Q_0, 20, 20 \rightarrow 0$

$Q \rightarrow 0.62$

~~$Q_0, 20, 20 \rightarrow 0$~~

$Q \rightarrow 0.62$

$Q_0, 0, 20 \rightarrow 0$

$Q \rightarrow 0.62$

$Q_0, 0, 20 \rightarrow 0$

$Q \rightarrow 0.62$

$Q_0, 0, 20 \rightarrow 0$

$Q \rightarrow 0.62$

$Q_0, 0, 20 \rightarrow 0$

$Q \rightarrow 0.62$

$Q_0, 0, 20 \rightarrow 0$

$Q_0, 20, 20 \rightarrow 0$

Convert the following given PDA to CFG.

$$① \delta(q_0, a, z_0) = (q_0, xz_0)$$

$$② \delta(q_0, a, x) = (q_0, xx)$$

$$③ \delta(q_0, b, x) = (q_1, \epsilon)$$

$$④ \delta(q_1, b, x) = (q_1, \epsilon)$$

$$⑤ \delta(q_1, \epsilon, z_0) = (q_1, \epsilon)$$

a - push i.e. Stack symbol.
b - pop i.e. Not b is not stack symbol.

Soln.

$$M = \left\{ \underbrace{\{q_0, q_1\}}_Q, \underbrace{\{a, b\}}_\Sigma, \underbrace{\{z_0\}}_\Gamma, \delta, q_0, z_0, F \right\}$$

$$G = (V, T, P, S)$$

Variable $V = (S, (q_0, a, q_0) (q_0, a, q_1) (q_1, a, q_0) (q_1, a, q_1), (q_0, z_0, q_0) (q_0, z_0, q_1) (q_1, z_0, q_0) (q_1, z_0, q_1))$
 $(Q \times \Gamma) \rightarrow$ initial state.

$$T \rightarrow (a, b)$$

$$S \rightarrow [q_0, z_0, q_0] \mid [q_0, z_0, q_1]$$

δ - become.

① $\delta(q_0, a, z_0) \rightarrow (q_0, xz_0)$

$[q_0, z_0, q_0]$	$\rightarrow$	q	$[q_0, x, q_0]$	$[q_0, z_0, q_0]$
$[q_0, z_0, q_0]$	$\rightarrow$	q	$[q_0, x, q_1]$	$[q_1, z_0, q_0]$
$[q_0, z_0, q_1]$	$\rightarrow$	q	$[q_0, x, q_0]$	$[q_0, z_0, q_1]$
$[q_0, z_0, q_1]$	$\rightarrow$	q	$[q_0, x, q_1]$	$[q_1, z_0, q_1]$

$2^2 = 4$ Row & 2 Column on R.H.S.

② $\delta(q_0, a, x) \rightarrow (q_0, xx)$

$2^2 = 4$ Row & 2 Column on R.H.S.

$[q_0, x, q_0]$	$\rightarrow$	q	$[q_0, x, q_0]$	$[q_0, x, q_0]$
$[q_0, x, q_0]$	$\rightarrow$	q	$[q_0, x, q_1]$	$[q_1, x, q_0]$
$[q_0, x, q_1]$	$\rightarrow$	q	$[q_0, x, q_0]$	$[q_0, x, q_1]$
$[q_0, x, q_1]$	$\rightarrow$	q	$[q_0, x, q_1]$	$[q_1, x, q_1]$

$$(3) \delta(q_0, b, x) \rightarrow (q_1, \epsilon)$$

$q^0 = 1$ Row 1.0 column on R.H.S.

$$[\underline{q_0, x, q_1}] \rightarrow b$$

$$(4) \delta(q_1, b, x) \rightarrow (q_1, \epsilon)$$

$$[\underline{q_1, x, q_1}] \rightarrow b$$

Remember these are in square bracket.

$$(5) \delta(q_1, \epsilon, z_0) \rightarrow (q_1, \epsilon)$$

$$[\underline{q_1, z_0, q_1}] \rightarrow \epsilon$$

So our grammar become.

$$S \rightarrow aAB | aCb | aAE | aCH$$

$$A \rightarrow aAA | aCF$$

$$C \rightarrow aAC | aCG | b$$

$$G \rightarrow b$$

$$E \rightarrow \epsilon$$